

An Analysis of Stable Diffusion

Stable Diffusion

$$\mathcal{L} = \mathbb{E}_{x, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

Stable Diffusion -- Part 1

Training data Stable Diffusion was trained on pairs of images and captions taken from LAION-5B, a publicly available dataset derived from Common Crawl data scraped from the web, where 5 billion image-text pairs were classified based on language and filtered into separate datasets by resolution, a predicted likelihood of containing a watermark, and predicted "aesthetic" score (e.g. subjective visual quality). The dataset was created by LAION, a German non-profit which receives funding from Stability AI. The Stable Diffusion model was trained on three subsets of LAION-5B: laion2B-en, laion-high-resolution, and laion-aesthetics v2 5+. A third-party analysis of the model's training data identified that out of a smaller subset of 12 million images taken from the original wider dataset used, approximately 47% of the sample size of images came from 100 different domains, with Pinterest taking up 8.5% of the subset, followed by websites such as WordPress, Blogspot, Flickr, DeviantArt and Wikimedia Commons. An investigation by Bayerischer Rundfunk showed that LAION's datasets, hosted on Hugging Face, contain large amounts of private and sensitive data.

Stable Diffusion -- Part 2

Stable Diffusion is a deep learning, text-to-image model released in 2022 based on diffusion techniques. The generative artificial intelligence technology is the premier product of Stability AI and is considered to be a part of the ongoing AI boom. It is primarily used to generate detailed images conditioned on text descriptions, though it can also be applied to other tasks such as inpainting, outpainting, and generating image-to-image translations guided by a text prompt. Its development involved researchers from the CompVis Group at LMU Munich and Runway with a computational donation from Stability and training data from non-profit organizations. Stable Diffusion is a latent diffusion model, a kind of deep generative artificial neural network. Its code and model weights have been released publicly, and an optimized version can run on most consumer hardware equipped with a modest GPU with as little as 2.4 GB VRAM. This marked a departure from previous proprietary text-to-image models such as DALL-E and Midjourney which were accessible only via cloud services.

Stable Diffusion -- Part 3

The text to image sampling script within Stable Diffusion, known as "txt2img", consumes a text prompt in addition to assorted option parameters covering sampling types, output image dimensions, and seed values. The script outputs an image file based on the model's interpretation of the prompt. Generated images are tagged with an invisible digital watermark to allow users to identify an image as generated by Stable Diffusion, although this watermark loses its efficacy if the image is resized or rotated. Each txt2img generation will involve a specific seed value which affects the output image. Users may opt to randomize the seed in order to explore different generated outputs, or use the same seed to obtain the same image output as a previously generated image. Users are also able to adjust the number of inference steps for the sampler; a higher value takes a longer duration of time, however a smaller value may result in visual defects. Another configurable option, the classifier-free guidance scale value, allows the user to adjust how closely the output image adheres to the prompt. More experimentative use cases may opt for a lower scale value, while use cases aiming for more specific outputs may use a higher value. Additional txt2img features are provided by front-end implementations of Stable Diffusion, which allow users to modify the weight given to specific parts of the text prompt. Emphasis markers allow users to add or reduce emphasis to keywords by enclosing them with brackets. An alternative method of adjusting weight to parts of the prompt are "negative prompts". Negative prompts are a feature included in some front-end implementations, including Stability AI's own DreamStudio cloud service, and allow the user to specify prompts which the model should avoid during image generation. The specified prompts may be undesirable image features that would otherwise be present within image outputs due to the positive prompts provided by the user, or due to how the model was originally trained, with mangled human hands being a common example.

Stable Diffusion -- Part 4

An "embedding" can be trained from a collection of user-provided images, and allows the model to generate visually similar images whenever the name of the embedding is used within a generation prompt. Embeddings are based on the "textual inversion" concept developed by researchers from Tel Aviv University in 2022 with support from Nvidia, where vector representations for specific tokens used by the model's text encoder are linked to new pseudo-words. Embeddings can be used to reduce biases within the original model, or mimic visual styles. A "hypernetwork" is a small pretrained neural network that is applied to various points within a larger neural network, and refers to the technique created by NovelAI developer Kurumuz in 2021, originally intended for text-generation

An Analysis of Stable Diffusion

Stable Diffusion

transformer models. Hypernetworks steer results towards a particular direction, allowing Stable Diffusion-based models to imitate the art style of specific artists, even if the artist is not recognised by the original model; they process the image by finding key areas of importance such as hair and eyes, and then patch these areas in secondary latent space. DreamBooth is a deep learning generation model developed by researchers from Google Research and Boston University in 2022 which can fine-tune the model to generate precise, personalised outputs that depict a specific subject, following training via a set of images which depict the subject.

original input image to be maintained in the generated output.

Stable Diffusion -- Part 5

Stable Diffusion also includes another sampling script, "img2img", which consumes a text prompt, path to an existing image, and strength value between 0.0 and 1.0. The script outputs a new image based on the original image that also features elements provided within the text prompt. The strength value denotes the amount of noise added to the output image. A higher strength value produces more variation within the image but may produce an image that is not semantically consistent with the prompt provided. There are different methods for performing img2img. The main method is SDEdit, which first adds noise to an image, then denoises it as usual in text2img. The ability of img2img to add noise to the original image makes it potentially useful for data anonymization and data augmentation, in which the visual features of image data are changed and anonymized. The same process may also be useful for image upscaling, in which the resolution of an image is increased, with more detail potentially being added to the image. Additionally, Stable Diffusion has been experimented with as a tool for image compression. Compared to JPEG and WebP, the recent methods used for image compression in Stable Diffusion face limitations in preserving small text and faces. Additional use-cases for image modification via img2img are offered by numerous front-end implementations of the Stable Diffusion model. Inpainting involves selectively modifying a portion of an existing image delineated by a user-provided layer mask, which fills the masked space with newly generated content based on the provided prompt. A dedicated model specifically fine-tuned for inpainting use-cases was created by Stability AI alongside the release of Stable Diffusion 2.0. Conversely, outpainting extends an image beyond its original dimensions, filling the previously empty space with content generated based on the provided prompt. A depth-guided model, named "depth2img", was introduced with the release of Stable Diffusion 2.0 on November 24, 2022; this model infers the depth of the provided input image, and generates a new output image based on both the text prompt and the depth information, which allows the coherence and depth of the